

HW 1.3 - Leaky Integrate-and-Fire Model

Governing Equation:

$$\tau \frac{dV}{dt} = -V + RI_0 \quad ; \quad V = u - u_{rest}$$

Reset condition: If $V \geq \theta$ (where $\theta = \vartheta - u_{rest}$), then $V \rightarrow 0$.

Interspike Interval (T):

Let $W = V - RI_0 \implies \tau \frac{dW}{dt} = -W$

$$W(t) = W_0 e^{-t/\tau}$$

$$V(t) - RI_0 = (V_0 - RI_0) e^{-t/\tau}$$

Assuming reset to rest ($V_0 = 0$):

$$V(t) = RI_0 (1 - e^{-t/\tau})$$

At $t = T$, $V(T) = \theta$:

$$\theta = RI_0 (1 - e^{-T/\tau})$$

$$e^{-T/\tau} = 1 - \frac{\theta}{RI_0}$$

$$T = -\tau \ln \left(1 - \frac{\theta}{RI_0} \right) \quad (RI_0 > \theta)$$

Firing Rate (f):

$$f = \frac{1}{T} = \frac{-1}{\tau \ln(1 - \frac{\theta}{RI_0})}$$

Plots:

